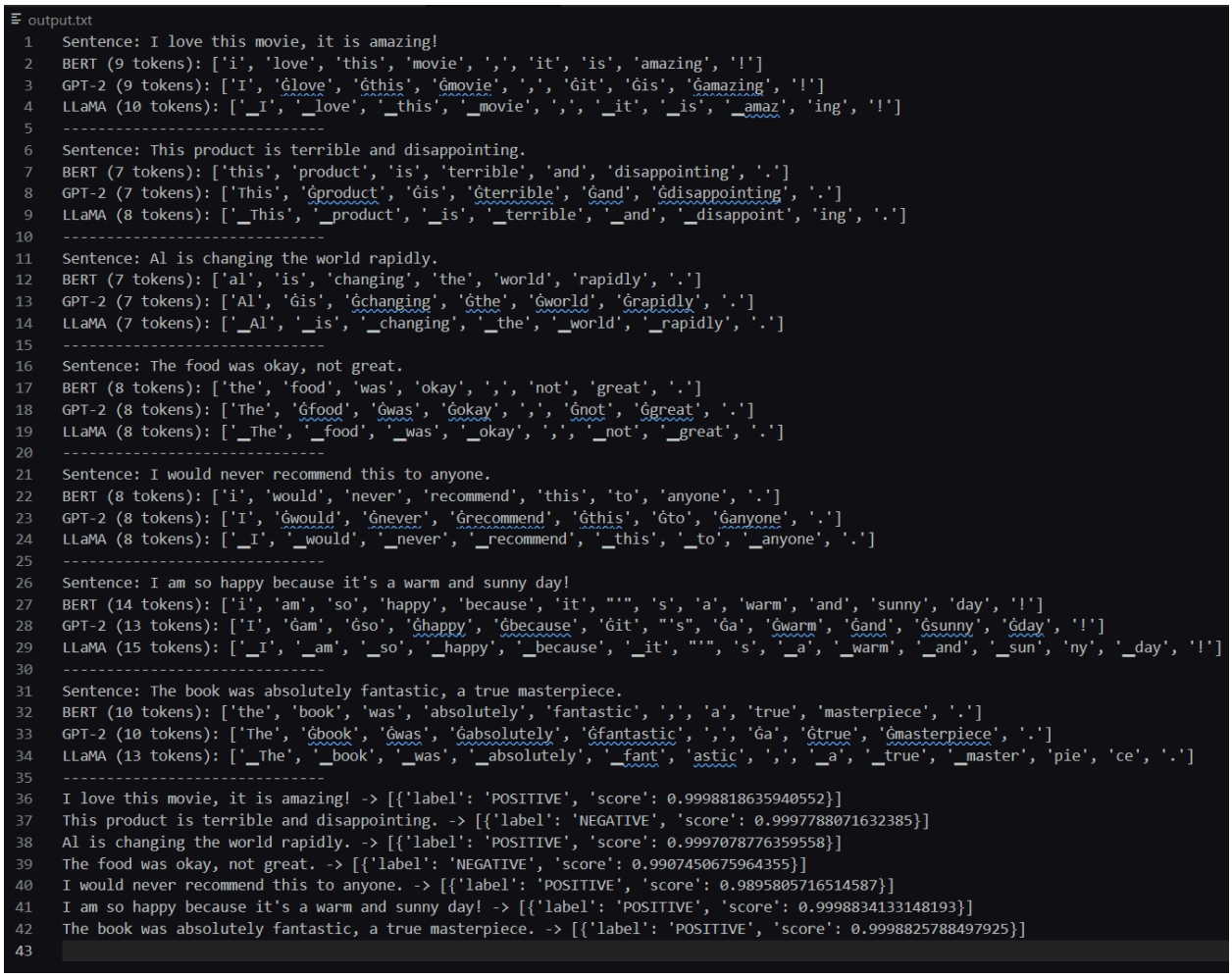
CS 3402 Homework 2

Part 2: Tokenization Analysis



output.txt

```
Sentence: I love this movie, it is amazing!
BERT (9 tokens): ['i', 'love', 'this', 'movie', ',', 'it', 'is', 'amazing', '!']
GPT-2 (9 tokens): ['I', 'Ġlove', 'Ġthis', 'Ġmovie', ',', 'Ġit', 'Ġis', 'Ġamazing', '!']
LLaMA (10 tokens): ['_I', '_love', '_this', '_movie', ',', '_it', '_is', '_amaz', 'ing', '!']
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Sentence: This product is terrible and disappointing.
BERT (7 tokens): ['this', 'product', 'is', 'terrible', 'and', 'disappointing', '.']
GPT-2 (7 tokens): ['This', 'Ġproduct', 'Ġis', 'Ġterrible', 'Ġand', 'Ġdisappointing', '.']
LLaMA (8 tokens): ['_This', '_product', '_is', '_terrible', '_and', '_disappoint', 'ing', '.']
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Sentence: Al is changing the world rapidly.
BERT (7 tokens): ['al', 'is', 'changing', 'the', 'world', 'rapidly', '.']
GPT-2 (7 tokens): ['Al', 'Ġis', 'Ġchanging', 'Ġthe', 'Ġworld', 'Ġrapidly', '.']
LLaMA (7 tokens): ['_Al', '_is', '_changing', '_the', '_world', '_rapidly', '.']
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Sentence: The food was okay, not great.
BERT (8 tokens): ['the', 'food', 'was', 'okay', ',', 'not', 'great', '.']
GPT-2 (8 tokens): ['The', 'Ġfood', 'Ġwas', 'Ġokay', ',', 'Ġnot', 'Ġgreat', '.']
LLaMA (8 tokens): ['_The', '_food', '_was', '_okay', ',', '_not', '_great', '.']
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Sentence: I would never recommend this to anyone.
BERT (8 tokens): ['i', 'would', 'never', 'recommend', 'this', 'to', 'anyone', '.']
GPT-2 (8 tokens): ['I', 'Ġwould', 'Ġnever', 'Ġrecommend', 'Ġthis', 'Ġto', 'Ġanyone', '.']
LLaMA (8 tokens): ['_I', '_would', '_never', '_recommend', '_this', '_to', '_anyone', '.']
------------------------------
Sentence: I am so happy because it's a warm and sunny day!
BERT (14 tokens): ['i', 'am', 'so', 'happy', 'because', 'it', "'", 's', 'a', 'warm', 'and', 'sunny', 'day', '!']
GPT-2 (13 tokens): ['I', 'Ġam', 'Ġso', 'Ġhappy', 'Ġbecause', 'Ġit', "'s", 'Ġa', 'Ġwarm', 'Ġand', 'Ġsunny', 'Ġday', '!']
LLaMA (15 tokens): ['_I', '_am', '_so', '_happy', '_because', '_it', "'", 's', '_a', '_warm', '_and', '_sun', 'ny', '_day', '!']
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Sentence: The book was absolutely fantastic, a true masterpiece.
BERT (10 tokens): ['the', 'book', 'was', 'absolutely', 'fantastic', ',', 'a', 'true', 'masterpiece', '.']
GPT-2 (10 tokens): ['The', 'Ġbook', 'Ġwas', 'Ġabsolutely', 'Ġfantastic', ',', 'Ġa', 'Ġtrue', 'Ġmasterpiece', '.']
LLaMA (13 tokens): ['_The', '_book', '_was', '_absolutely', '_fant', 'astic', ',', '_a', '_true', '_master', 'pie', 'ce', '.']
------------------------------
I love this movie, it is amazing! -> [{'label': 'POSITIVE', 'score': 0.9998818635940552}]
This product is terrible and disappointing. -> [{'label': 'NEGATIVE', 'score': 0.9997788071632385}]
Al is changing the world rapidly. -> [{'label': 'POSITIVE', 'score': 0.9997078776359558}]
The food was okay, not great. -> [{'label': 'NEGATIVE', 'score': 0.9907450675964355}]
I would never recommend this to anyone. -> [{'label': 'POSITIVE', 'score': 0.9895805716514587}]
I am so happy because it's a warm and sunny day! -> [{'label': 'POSITIVE', 'score': 0.9998834133148193}]
The book was absolutely fantastic, a true masterpiece. -> [{'label': 'POSITIVE', 'score': 0.9998825788497925}]
```

1. Which tokenizer produces the most tokens on average?

LLaMA produces the most tokens on average.

- BERT Total Tokens: 63 (Average: 9.0)
- GPT-2 Total Tokens: 62 (Average: ~8.86)
- LLaMA Total Tokens: 69 (Average: ~9.86)

2. Which tokenizer uses subword splitting (e.g., play ##ing)?

BERT is the tokenizer designed to use the ## prefix for subword splitting. While this output did not trigger this for BERT, the LLaMA tokenizer demonstrated significant subword splitting, breaking words like "amazing" into ['_amaz', 'ing'] and "disappointing" into ['_disappoint', 'ing'].

3. Briefly explain why different tokenizers produce different outputs.

Different tokenizers utilize different algorithms, such as WordPiece (BERT), Byte Pair Encoding (GPT-2), or SentencePiece (LLaMA), and are trained on different vocabulary sizes. A tokenizer with a smaller vocabulary or a different splitting logic will break the same sentence into a different number of sub-tokens to ensure every character sequence can be represented.

Part 3: Sentiment Analysis (Hugging Face)

4 & 5. Analysis of Results and Ambiguity

Sentence	Predicted Label	Score
"I love this movie, it is amazing!"	POSITIVE	0.9998
"This product is terrible and disappointing."	NEGATIVE	0.9997
"AI is changing the world rapidly."	POSITIVE	0.9997
"The food was okay, not great."	NEGATIVE	0.9907
"I would never recommend this to anyone."	POSITIVE	0.9895

"I am so happy because it's a warm and sunny day!"	POSITIVE	0.9998
"The book was absolutely fantastic..."	POSITIVE	0.9998

Sentences that are ambiguous or difficult:

- "I would never recommend this to anyone.": The model incorrectly labeled this as POSITIVE with high confidence (0.989). This is a failure because the model likely focused on the word "recommend" and failed to process the negation "never".
- "The food was okay, not great.": The model labeled this NEGATIVE. This is ambiguous because "okay" is neutral, but the model's binary choice (Positive/Negative) forced it into a negative classification.

Part 4: Sentiment Analysis with Gemini, ChatGPT, and Claude

Here are Gemini's classifications and explanations:

Sentence	Sentiment Label	Gemini's Brief Explanation
"I love this movie, it is amazing!"	Positive	Strong positive adjectives ("love," "amazing") indicate high satisfaction.
"This product is terrible and disappointing."	Negative	"Terrible" and "disappointing" are explicitly negative descriptors of quality.
"AI is changing the world rapidly."	Neutral	This is a factual statement about the pace of change without an inherent value judgment.

"The food was okay, not great."	Neutral/Negative	"Okay" suggests a baseline acceptability, while "not great" leans slightly negative.
"I would never recommend this to anyone."	Negative	This is a strong statement of dissatisfaction and a warning to others.
"I am so happy because it's a warm and sunny day!"	Positive	"Happy" is a direct emotional indicator of positive sentiment.
"The book was absolutely fantastic, a true masterpiece."	Positive	"Fantastic" and "masterpiece" represent the highest level of positive acclaim.

Here are ChatGPT’s classifications and explanations:

Sentence	Sentiment Label	ChatGPT’s Brief Explanation
"I love this movie, it is amazing!"	Positive	Strong positive words like “love” and “amazing” clearly express enthusiasm and approval.
"This product is terrible and disappointing."	Negative	Words like “terrible” and “disappointing” indicate strong dissatisfaction.
"AI is changing the world rapidly."	Neutral	This is a factual statement without clear positive or negative emotion.

"The food was okay, not great."	Neutral (slightly negative)	“Okay” suggests mild satisfaction, but “not great” adds a slightly negative tone.
"I would never recommend this to anyone."	Negative	The phrase “never recommend” shows a clear negative opinion.
"I am so happy because it's a warm and sunny day!"	Positive	Words like “so happy” and pleasant weather imagery convey a strong positive feeling.
"The book was absolutely fantastic, a true masterpiece."	Positive	Intensifiers like “absolutely” and praise like “masterpiece” signal very strong positive sentiment.

Here are Claude’s classifications and explanations:

Sentence	Sentiment Label	Claude’s Brief Explanation
"I love this movie, it is amazing!"	Positive	Strong enthusiasm from "love" and "amazing" signals clear, unambiguous positive sentiment.
"This product is terrible and disappointing."	Negative	"Terrible" and "disappointing" are strongly negative evaluative words with no hedging.

"AI is changing the world rapidly."	Neutral	A factual, descriptive statement – no positive or negative evaluative language is used.
"The food was okay, not great."	Negative	Mildly mixed – "okay" is lukewarm and "not great" undercuts it, resulting in a net neutral impression.
"I would never recommend this to anyone."	Negative	"Never recommend" is a strong refusal implying the experience was bad enough to actively warn others away.
"I am so happy because it's a warm and sunny day!"	Positive	"So happy," "warm," and "sunny" all carry positive emotional valence, amplified by the exclamation mark.
"The book was absolutely fantastic, a true masterpiece."	Positive	"Absolutely fantastic" and "true masterpiece" are superlative praise – among the strongest positive expressions possible.

Do different LLMs give the same predictions?

For the most part, yes. All three models (Gemini, ChatGPT, and Claude) agreed on the clear positive and negative sentences. The only variance occurred with "The food was okay, not great", where Gemini and ChatGPT leaned toward a "Neutral/Negative" or "Neutral" hybrid, while Claude ultimately labeled it "Negative" while noting its "net neutral impression" in the text

Which model provides the most reasonable explanation?

While subjective, Claude provides arguably the most nuanced linguistic analysis. It correctly identifies concepts like "hedging" and "emotional valence". Gemini is also very effective, specifically noting that the AI sentence is a "factual statement" without an inherent value judgment.

Compare LLM results with Hugging Face results. What differences do you observe?

There are two major differences:

Nuance vs. Binary: The Hugging Face pipeline is limited to a binary POSITIVE/NEGATIVE output, which forced neutral sentences like "AI is changing the world" into a positive label. The LLMs correctly identified these as Neutral.

Contextual Negation: The Hugging Face model failed significantly on "I would never recommend this to anyone," labeling it POSITIVE (0.989) likely because it focused on the keyword "recommend". All three LLMs correctly identified the sentiment as Negative by processing the negation "never".